Congruence modulo 2 relation, 488
Congruence modulo 3 relation, 500–501, 516–517
Congruence modulo *n*, 516–518, 526–528
Congruence relations, 516–518
Congruences, evaluating, 518
Conjecture, 179–180
Conjunctions, 41, 76
Connected components, 683
  matrices and, 702–703
Connected graphs, 682–683
Connectedness, 682–683
Consecutive integers, 203
Consequent, 54
Constant function, 21
Constructive proofs of existence, 164
Context-free languages, 828–829, 854
Contradiction
  definition of, 48
  indirect argument by, 218–225
  logical equivalence and, 49
  method of proof by, 218–221
  proof by, 223–224
  rule, 74–75, 76
  valid arguments and, 74–76
Contradictory statements, 48
Contraposition
  indirect argument by, 218–225
  method of proof by, 222–224
Contrapositive
  of conditional statement, 57–58
  of universal conditional statements, 127–128
  writing, 58
Converse
  of conditional statement, 58–59
  of universal conditional statements, 127–128
Converse error, 72, 73, 153, 155–156
Corollary, 187
Correctness of post-condition, 317
Countable sets, 475–477
Counterexamples, 110
  direct proof and I, 161–171
  direct proof and II (writing advice), 173–180
  direct proof and III (rational numbers), 183–187
  direct proof and IV (divisibility), 190–197
  direct proof and V (division into cases and the quotient-remainder theorem), 200–208
  direct proof and VI (floor and ceiling), 211–216
  disproving universal statements by, 164–165
  divisibility and, 194–195
  for set identity, 407–409
Counting, 564–676
  advice about, 629–630
  double, 629–630
  elements in Cartesian product, 576–577
  elements in intersection, 597–599
  elements of disjoint sets, 589–599
  elements of general union, 596–597
  elements of list, 568–570
  input/output tables for circuit with two input signals, 577–578
  integral solutions of equation, 638–639
  Internet addresses, 594–595
  iterations of loop, 637–638
  iterations of nested loop, 578
  number of integers divisible by 5, 589–590
  Pascal's formula and, 642–646
  passwords, 592–593
  permutations, 580–584
  PINs, 575–576
  PINs, with repeated symbols, 590–591
  possibility trees and multiplication rule, 573–584
  probability and, 564–570
  Python identifiers, 593–594
  subsets of a set, 617–630
  triples, 637
  walks of length *N*, 707–710
Critical Path Method (CPM), 558–560
Cross products, 519
Cryptographic hash functions, 445
Cryptography, 524–526
  Caesar cipher, 524–525
  definition of, 524
  message authentication, 542–543
  public-key, 525–526
  RSA cipher, 525–526, 536–539
  RSA cryptography, 540–543
  uses of, 524

Databases, relational, 491–492
Data type, 244
Dates, regular expression for, 838
Da Vinci, Leonardo, 1
Davis, Philip J., 211
Day of week, computing, 202
Debate, 218
Decimal expansions, of fractions, 608–610
Decimal notation, 93
  conversion of binary notation to and from, 95, 271–273
  converting from hexadecimal notation to, 104–105
Decimals, repeating, 184
Decision trees, 721–722
Decoding functions, 431
Decreasing functions, 765–766
Decryption, 524–525
Decryption key, 537
Dedekind, Richard, 520
Deductive reasoning, 275
  rules of inference and, 69–72
  universal instantiation and, 146–147
Degree of vertex, 31–32
De Grey, Aubrey, 34
De Morgan, Augustus, 37, 46, 277
De Morgan's laws, 45–46, 125, 395
  applying, 46–47
  Boolean algebra and, 416

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

De Morgan's laws (*Continued*)
cautionary examples of, 47–48
definition of, 46
inequalities and, 47
for sets, proof of, 398–400
Descartes, René, 131, 760
Descendant, 733
Diaconis, Persi, 566
Diagrams
See also Arrow diagrams
Hasse, 550–553
testing for validity using, 151–155
transition, 843–844
tree, 663–665
Dice rolls, 567
Difference rule, 590–595
Differences of sets, 381, 382
Digital computers, 842–843
Digital logic circuits, 79–90
background on, 79–81
black boxes and gates, 81–83
Boolean expressions and, 84–87
combinational, simplifying, 88–89
equivalence classes of, 515–516
equivalence of, 509
equivalent, 89
input/output table for, 83–84, 87–88
NAND and NOR gates, 89–90
Digraphs, 29
Dijkstra, Edsger W., 314, 315, 377, 752
Dijkstra's shortest path algorithm, 751–757
Dirac, P. A. M., 495
Directed edges, 29
Directed graphs, 29, 495
definition of, 29
matrices and, 699–701
of partial order relation, 552–553
of relation, 490–491
Direct proof
counterexample I and, 161–171
counterexample II and (writing advice), 173–180
counterexample III and (rational numbers), 183–187
counterexample IV and (divisibility), 190–197
counterexample V and (division into cases and the quotient-remainder theorem), 200–208
counterexample VI and (floor and ceiling), 211–216
method of, 167
of theorem, 168–170
Dirichlet, J. P. G. L., 604
Dirichlet, Lejeune, 426
Dirichlet box principle. *See* Pigeonhole principle
Discovery, 161
Discrete mathematics, 8
Disjoint events, 668–669
Disjoint sets, 384–385
counting elements of, 589–599
Disjunction, 41–42
Disjunctive normal form, 88
Disproof
of alleged property of floor, 213
of alleged set property, 407–409
by counterexample, 164–165
of existential statements, 179
Disquisitiones Arithmeticae (Gauss), 518
Distinct equivalence classes, 511–517
Distinct-roots theorem, 353–360
Distributive laws, 394, 416, A–1
general, 402–403
proof of, 395–398
div, 201–202
computing, 216
Divide-and-conquer strategy, 813–814
"Divides" relations
on set of positive integers, 548–549
testing for antisymmetry of, 547–548
Divisibility, 190–197
algebraic expressions and, 191–192
checking for nondivisibility, 192
checking proposed properties, 195
concept of, 190
counterexamples and, 194–195
by a prime, 302–303
by prime numbers, 194
prime numbers and, 192
properties of, 191
proving, with mathematical induction, 291–293
proving properties of, 192–194
transitivity of, 192–194
unique factorization of integers theorem and, 195–197
Division algorithm, 248–250
correctness of, 319–321
Division into cases, proof by, 204–205
Divisors
greatest common, 250–252, 253, 532–533
of one, 191
positive, of positive integer, 191
of zero, 190–191
Domains, 16, 425
Dot product, 703
Double complement laws, 394, 416
proof of, 417–418
Double counting, 629–630
Double negative property, 44–45
Dual identity, 417
Duality principle, of Boolean algebra, 417
Dummy variables, 266
in loop, 270–271

EBCDIC (Extended Binary-Coded Decimal Interchange Code), 483
Edge-endpoint functions, 25
Edges, 24–25, 29, 681
Edinburgh Prolog, 141*n*
Edison, Thomas Alva, 352
8-bit two's complements, 99–103
Einstein, Albert, 589

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Electrical engineering, 828
Electronic technology, 80–81
Element arguments, 391
Elementary operations, 788–789
Elements
  in Cartesian products, 576–577
  comparable, 553
  of disjoint sets, counting, 589–590
  of elements, 628–629
  of general union, counting, 596–597
  greatest, 554–555
  in intersection, counting, 597–599
  least, 308–309, 554–555
  of list, counting, 568–570
  maximal, 554–555
  minimal, 554–555
  noncomparable, 553
  ordered selection of, 618
  unordered selection of, 618
Elements of Geometry (Euclid), 229, 230
Elimination, 70, 76
Elkies, Noam, 180
Ellipsis, 7
Emacs, 836
Empty sets, 384, 401–403
  proving set is, 402
  as subset of every set, 401–402
  uniqueness of, 401
Encoding functions, 431
Encryption, 524–525
  See also Cryptography
Endpoints, 25, 29
Enumeration, complete, 618
Equality
  of functions, 428
  properties of, 498–499
  proving, 284–285
  set, 379–380
  test for function, 428
Equally likely probability formula, 566
Equations, number of integral solutions of, 638–639
Equivalence, modular, 526–527
Equivalence classes
  of congruence modulo 3, 516–517
  definition of, 510
  of digital logic circuits, 515–516
  of equivalence relation, 510–516
  of identifiers, 512
  of identity relation, 512–515
  rational numbers as, 519–520
  of relation given set of ordered pairs, 511
  of relation on set of subsets, 511–512
  star *-equivalence classes, 863
Equivalence of states, 860
Equivalence relations, 505–520
  congruence modulo *n*, 527–528
  definition of, 508–510
  digital logic circuits and, 509
  equivalence classes of, 510–516
  graph isomorphisms as, 715
  partition inducing, 506–508, 514–515
  on set of subsets, 508–509
Equivalent automata, 866–868
Equivalent circuits, 89
Equivalent forms, of universal and existential statements, 114–115
Equivalent states, 859–863
Eratosthenes, 227
Escape characters, 833
Etchemendy, John, 117–118
Euclid, 196, 229, 230
Euclidean algorithm, 250–254
  correctness of, 321–323
  extending, 531–533
Euclid's lemma, 539–540
Euler, Leonhard, 180, 677
Euler circuits, 684–689
  definition of, 684
  finding, 687–688
  vs. Hamiltonian circuits, 690
  showing graph does not have, 685–687
Euler trails, 688–689
Even integers, 161–163
  Goldbach's conjecture about, 180
  properties of, 186–187
Even parity, 835
Event(s), 566
  disjoint, 668–669
  of events, 668–672
  independent, 668–672
  probability of complement of, 591, 656
  probability of general union of two, 657–658
Eventual falsity of guard, 317
Eventual-state function, 846–847
Examples, arguing from, 176
Exclusive *or*, 42–43
Exhaustion, method of, 111, 166
Existence
  constructive proofs of, 164
  of graphs, 236–237
  of identity elements, A–1
  nonconstructive proof of, 164
Existential instantiation, 148, 168
Existential quantifiers, 111–112
Existential statements, 2
  definition of, 111
  disproving, 179
  equivalent forms of, 114–115
  proving, 164
  truth and falsity of, 112
  universal, 3–5
Existential universal statements
  definition of, 4
  rewriting, 5
Expanded form, changing between summation notation and, 261–262

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Expected value, 658–660, 670
Expert systems, 156
Explicit formula
for Fibonacci sequence, 358–360
to fit given initial term, 260–261
for geometric sequence, 343
incorrect, 348–349
for method of iteration, 340–342
for sequences, 259–261
simplifying solutions obtained by iteration, 345–347
for Tower of Hanoi, 345–346
Exponential functions
with base *b*, 449–450
graphs of, 800–801
one-to-oneness of, 450–451
relations between logarithmic and, 449–451
Exponential notation, 93
Exponential orders, 806–811, 824
Exponents, law of, 449
Expressions
See also Regular expressions
in algorithmic language, 244
numerical, 341
Extended Binary-Coded Decimal Interchange Code (EBCDIC), 483
Extended Euclidean algorithm, 531–533
Extension, axiom of, 7

Factorial notation, 268–270
Factorials, computing with, 269–271
Factorization, unique factorization of integers theorem, 195–197
Fallacies, 72–74
Fallacy of affirming the consequent, 73
Fallacy of denying the antecedent, 73–74
False hypotheses, conditional statements with, 54–55
False negatives, 667
False positives, 667
Fantasy rule for mathematical proof, 393
Fermat, Pierre de, 179–180, 190, 277, 568
Fermat primes, 232
Fermat's last theorem, 232
Fermat's little theorem, 540–541
Fibonacci (Leonardo of Pisa), 332
Fibonacci numbers, 332–333
Fibonacci sequence, 358–360
Field axioms, A–1, A–2
Final term, 259
Finite relations
inverse of, 489
testing for antisymmetry of, 547
Finite sequences, 270
Finite sets, 473
composition of functions defined by, 463
functions and relations on, 19–20
one-to-one functions defined on, 441
onto functions defined on, 446–447
pigeonhole principle and, 612–614
properties of relations on, 496–498
Finite-state automata, 836, 841–855
definition of, 843–845
designing, 847–849
digital computers, 842–843
equivalence states and, 859–863
eventual-state function and, 846–847
given by annotated next-state table, 844–845
given by transition diagram, 843–844
language accepted by, 845–846
nondeterministic, 853
regular expressions and, 851–853
simplifying, 858–868
software simulating, 849–851
First-order logic, language of, 141
Floor, 211–216
Floor function, 304–305, 762–763, 791–793
Floyd, Robert W., 315
Forests, 720
Formal language(s), 828, 829–832
examples of, 830
vs. informal, 112–113
translating informal to, 134–136
Formal logical notation, 139–141
Formulas
See also Explicit formula
checking correctness of, using mathematical induction, 347–348
composition of functions defined by, 462
computational, 620
deciding on, 639
Pascal's formula, 642–646
for probability of complement of event, 591
proving, 275–286
summation of first *n* integers, 278–282
For-next loops, 245–246, 793–794
dummy variable in, 270–271
trace table for, 247–248
Forster, E. M., 79
Forward chaining, 398
Four-color theorem, 34
Four-function calculator, computing div and mod with, 201–202
Fractional part, 213
Fractions, decimal expansions of, 608–610
Frege, F. L. G., 520
Frege, Gottlob, 110
Friedl, Jeffrey E. F., 851*n*
Frye, Roger, 180
Full-adders, 97–99
Full binary tree, 734
Fuller, R. Buckminster, 713
Function notation, 19
Functions, 425–486
See also Composition of functions; Exponential functions; Finite-state automata; Logarithmic functions
absolute value, 765

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Ackermann, 372–373
acting on sets, 434–435
arrow diagrams of, 426–427
Boolean, 432–433, 828
cardinality of, 483–484
Cartesian product defining, 430
composition of, 461–470
constant, 21
defined on sets of integers, graphing, 763
definition of, 425–426
domain, 425
edge-endpoint, 25
encoding/decoding, 431
equality of, 21, 428
eventual-state, 846–847
examples of, 429–432
exponential, 449–451, 800–801
on finite sets, 19–20
floor, 762–763, 791–793
formulas defining, 21
general sets defining, 425–435
Hamming distance, 431–432
hash, 443–445
identity, 463–464
identity, on X, 429
increasing and decreasing, 765–766
inverse, 440, 454–457, 464–466
language of, 18–20
logarithmic, 430–431, 449–451, 800, 801–803
machines, 20–21
McCarthy's 91, 372
multiple of, graph of, 764
next-state, 843
noncomputable, 483
nonfunctions and, 427
not well-defined, 433–434
one-to-one, 439–445, 613–614
one-to-one correspondences and, 451–454
one-way, 445
onto, 445–449, 468–470, 613–614
polynomial, 773–783
power, 761–762, 773
power sets defining, 429–430
from power set to set of strings, 452
probability, 655
propositional, 108
range, 426
real-valued, of real variable, 760–766
recursive, 372–374
sequences and, 429
squaring, 21
string-reversing, 452–453
successor, 21
test for equality, 428
of two variables, 453–454
with union, 434–435
well-defined, 433–434
Fundamental theorem of arithmetic, 195–197
Galilei, Galileo, 473
Gambling, 659–660
Gates, 81–83
Gauss, Carl Friedrich, 173, 196, 201, 281, 518
gcd. *See* Greatest common divisor
General formula, for sequence, 259
Generalizations, 69–70, 76
  from the generic particular, 166–167, 184–185
  universal, 166
Generalized associative law, deriving, 412
Generalized distributive law, 402–403
General recursive definitions, 364–372
Generic particular, generalizing from, 166–167, 184–185
Geometric sequence, 344–345
  explicit formula for, 343
  sum of, 282–284, 285–286
Germain, Marie-Sophie, 232
Gibbs, Josiah Willard, 15
Gilbert, William S., 642
Gleick, James, 180
Gödel, Kurt, 420
Goldbach, Christian, 180
Goldbach conjecture, 180
Golden ratio, 360*n*, 364
Golomb, Solomon, 296
Grammar rules, 722–723
Grammars, 828
Graph(s), 677–720
  See also Directed graphs; Tree(s)
  of absolute value function, 765
  acquaintance, 237
  bipartite, 243
  circuit-free, 720
  circuits and, 677–692, 730
  coloring maps with, 32–34
  complete, 240–241, 626
  complete bipartite, 241, 626
  connected, 682–683
  definition of, 25, 679–680
  degree of vertices, 31–32
  directed, 29, 699–701
  directed, of relation, 490–491, 495
  disconnected, 682–683
  edges, 24–25
  examples of, 28–34
  existence of, determining, 236–237
  of exponential functions, 800–801
  of floor function, 762–763
  forest, 720
  of function *f*, 760–761
  of functions defined on sets of integers, 763
  isomorphisms of, 713–718
  language of, 24–34
  of logarithmic functions, 801–803
  matrix representations of, 698–710
  of multiple of a function, 764

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Graph(s) (*Continued*)
paths in, 677–692
pictorial representation of, 26–28
of power functions, 761–762
problem-solving using, 30–31
of real-valued functions of a real variable, 760–766
to represent knowledge, 29–30
to represent World Wide Web, 28–29
simple, 239–240, 717–718
subgraphs, 681–682
terminology, 25–26
total degree of, 235–236
trails and, 677–692
tree, 720, 727
uses of, 677
vertices, 24–25
vertices with odd degree, 238
walks in, 707–710
weighted, 745–748
Graph theory, origins of, 677–679
Greatest common divisor (gcd), 250–252, 253
writing as linear combination, 532–533
Greatest elements, 554–555
Green, Ben Joseph, 232
grep, 836
Gries, David, 315
Griggs, Jerrold, 394
Guard, 245, 316, 319
Guard condition, 320
eventual falsity of, 320, 322

Haken, Wolfgang, 34
Half-adders, 97, 99
Hall, Monty, 567
Halting problem, 420–421
Hamilton, Sir William, 689
Hamiltonian circuits, 689–692
definition of, 690
vs. Euler circuits, 690
showing graph does not have, 691–692
traveling salesman problem and, 692
Hamming, Richard W., 431–432
Hamming distance function, 431–432
Handshake theorem, 235–241
Hardy, G. H., 218, 258, 524, 543
Harmonic sums, 809–810
Hash functions, 443–445
computing values of, 444–445
cryptographic, 445
Hasse, Helmut, 551
Hasse diagrams, 550–553
Hausdorff, Felix, 10
Hersh, Reuben, 211
Hexadecimal notation, 103–104
bination notation converting from/to, 105–106
decimal notation converted from, 104–105
High-level computer languages, 828–829
Hilbert, David, 414, 843
Hofstadter, Douglas, 364, 367, 391, 393
Horizontal axes, 760
Hydrocarbon molecules, structure of, 723–724
Hypertext transfer protocols (HTTPs), 28
Hypothesis, 53, 54, 62, 66
false, 54–55

Idempotent laws, 394
proof of, 418
Identifiers
equivalence classes of, 512
Python, counting, 593–594
relation on set of, 510
Identities, set, 394–400
Identity elements, A–1
Identity function, 463–464
Identity function on X, 429
Identity laws, 394, 416
Identity matrices, 706–707
Identity relations, equivalence classes of, 512–515
If, 162
misuse of, 177–178
If-then-else statements, 204, 245, 793–794
If-then statements
converting necessary condition to, 62
converting only if to, 59–60
converting sufficient condition to, 61
execution of, 245
negation of, 57
representation as *or*, 56
Implicit quantification, 116–117
Inclusion/exclusion rule, 595–599
Increasing functions, 765–766
Independent events, 668–672
disjoint events and, 668–669
probability and, 669–670
three events, 671
Index, 259
Indirect argument
contradiction and contraposition, 218–225
famous theorems, 228–233
Indirect proof, when to use, 231–232
Induction, 275
See also Mathematical induction
structural, 364, 367–372
Inductive hypothesis, 279
Inductive property, 317
Inductive step, 278, 410
Inequalities
De Morgan's laws and, 47
notation for, 40
proving, with mathematical induction, 293–295
triangle, 207–208
Inference, rules of, 69–72, 76
Infinite relations, inverse of, 490
Infinite sequences, 259
Infinite sets, 473, 612
Cantor diagonalization process and, 477–482

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

cardinality and, 474–475
countability of, 475–477
one-to-one functions on, 441–443
onto functions on, 447–449
properties of relations on, 498–501
uncountable, 476
Infinitude of prime numbers, 230–231
Infix notation, 831
Informal language
vs. formal, 112–113
translating to formal, 134–136
Initial conditions, for recurrence relations, 325
Initial state, 843
Initial term(s), 259
finding explicit formula to fit, 260–261
Injective (one-to-one) functions, 440–443
Input alphabet, 843
Input arrays, length of, 817
Input/output table(s), 81
for circuit, 83–84, 87–88
for circuit with two input signals, 577–578
Input signals, 81
Insertion sort algorithm, 793–796
Instantiation
existential, 148, 168
universal, 146–147, 150
Integers, 8
addition/subtraction in two's complement form, 102–103
binary notation for, 94–95
binary representation of, 307–308
composite, 163–164
computer representation of, 99–101
consecutive, 203
divisibility of, 195–197
divisible by 5, counting number of, 589–590
even, 161–163, 186–187
graphing functions defined on sets of, 763
linear combination of, 532
mod 4, 204–205
odd, 161–163, 186–187
odd minus even, 178–179
pairwise relatively prime, 534
parity of, 203–204
positive divisor of positive, 191
prime, 163–164
properties of, 186–187
property of the set of, 368–369
quotient-remainder theorem and, 205
relatively prime, 534–535
representation of, 203–208
selecting pair of, with certain sum, 607–608
smallest positive, 135
square of, 222–223
square of odd, 205–207
in standard factored form, 196
unique factorization of integers theorem, 195–197
well-ordering principle for, 258, 308–310
Integral solutions of equation, 638–639
Intel 4004, 80
Internal vertex, 725–727
Internet Protocol (IP) addresses, 594–595
Intersection
counting elements of, 597–599
of sets, 381, 382, 383–384
with subset, 400
Interval notation, 382–383
Intervals, 382–383
Intractable problems, 824
Invalid arguments. *See* Arguments, valid and invalid
Invalidity, using diagrams to show, 152–153
Invariant for graph isomorphism, 716
Inverse
of conditional statement, 58–59
of relation, 488–490
of universal conditional statements, 127–128
Inverse error, 72, 73–74, 153, 155–156
Inverse functions, 440, 454–457, 464–466
Inverse modulo n, 534–536
Inverter, 82
Irrational numbers
determining rational numbers vs., 183–184
irrationality of square root of two, 228–230
sum or rational and, 220–221
Isbell, John, 34
Isolated vertex, 25
Isomers, 723
Isomorphic invariants, 716
Isomorphic structures, 867–868
Isomorphisms
definition of, 713
of graphs, 713–718
for simple graphs, 717–718
Iterations
counting, of loop, 578, 637–638
formulas to simplify solutions obtained by, 345–347
method of, 340–345
number of, depending on floor function, 791–793
solving recurrence relations by, 340–349
Iterative statements, 245–246

Java language, 201, 244, 836
Job scheduling problem, 558–560
Jumping to conclusions, 72

Kant, Immanuel, 37, 742
k-equivalence of states, 860–863
Keys, binary search trees, 739
Killian, Charles, 394
Kirchhoff, Gustav, 723
Kleene, Stephen C., 828, 851
Kleene closure of L, 832
Kleene closure of r, 832
Kleene closure of Σ, 830
Kleene's theorem, 851–853
Knuth, Donald, 169, 769, 787
Kolmogorov, Andrei Nikolaevich, 655

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Königsberg bridges puzzle, 677–679
Kronecker, Leopold, 706
Kruskal, Joseph, 745
Kruskal's algorithm, 745–748
Kuratowski, Kazimierz, 10

The Language of First-Order Logic (Barwise and Etchemendy), 117–118
Language(s)
  See also Computer languages; Formal language; Informal language
  accepted by automaton, 845–846
  ambiguous, 136–137
  context-free, 828–829, 854
  of first-order logic, 141
  formal vs. informal, 112–113
  new, from old, 832
  nonregular, 854–855
  regular, 854–855
  regular expression defining, 832–836
  translating from informal to formal, 134–136
Laplace, Pierre-Simon, 568, 655, 662
Law of exponents, 449
lcm. *See* Least common multiple
Leaf, 725–727
Least common multiple (lcm), 256
Least elements, 554–555
  finding, 308–309
Least nonnegative residues modulo *n*, 527
Least upper bound axiom, A–3
Left child, 734
Leibniz, Gottfried Wilhelm, 37, 151
Lemma, 207
  Euclid's, 539–540
Length
  of string, 369–370
  of walk, 707–710
"Less than or equal to" relations, 549
Less-than relations, 487, 499
Lexicographic order, 549–553
Linear combinations
  of relatively prime integers, 535
  writing greatest common divisor as, 532–533
Linear probe, 444
Linguistics, 723, 828
Lists, counting elements of, 568–570
Lobachevsky, Nicolai Ivanovitch, 546
Local call string, 857
Logarithmic functions, 800
  with base *b*, 430–431
  graphs of, 801–803
  orders involving, 807–808
  relations between exponential and, 449–451
Logarithmic orders, 806–811
Logarithms
  common, 451
  computing, with base 2 on calculator, 451
  natural, 451
  of numbers between two consecutive powers of 2, 802–803
  properties of, 450
  recurrence relations solved with, 804–806
Logic, 828
  of compound statements, 37–107
  De Morgan's laws of, 45–48
  digital logic circuits, 79–90
  form of argument, 37–38
  language of first-order, 141
  of quantified statements, 108–152
  symbolic, 37
Logical equivalence, 43–48
  conditional statements and, 55–56, 62
  contradictions, 49
  De Morgan's laws and, 45–48
  double negative property and, 44–45
  nonequivalence and, 45
  for quantified statements, 122–123, 125–126
  set properties and, 414–415
  statements/statement forms and, 43–44, 50
  summary of, 49–50
  tautologies, 49
Logical form, of arguments, 37–38
Logical operators, order of operations for, 60
Loop invariants, 316–319
Loop invariant theorem, 317
Loops, 25
  correctness of, to compute product, 318–319
  counting iterations of, 637–638
  dummy variable in, 270–271
  nested, 578, 790–791
  for-next, 245–248, 270–271, 793–794
  number of iterations of, 578
  pre-conditions and post-conditions, 316–317, 319
  while, 245–247, 316, 793–794
Lottery, expected value of, 658–659
Lovelace, Ada Augusta, Countess of, 244, 813
Lucas, Édouard, 328
Lukasiewicz, Jan, 831
Lynch, John, 180*n*

Mach, Ernst, 487
Main diagonal, of matrix, 698–699
Major premise, 68
Manin, I., 275
Maps, coloring, using graphs, 32–34
Mastering Regular Expressions (Friedl), 851*n*
Mathematical Analysis of Logic, The (Boole), 415
Mathematical Experience, The (Davis and Hersh), 211
Mathematical induction, 258, 275–297
  See also Well-ordering principle
  additional formals, deducing, 285–286
  applications, 289–297
  checking correctness of formula by, 347–348
  definition of, 275–277
  divisibility property proven with, 291–293
  equality proven with, 284–285
  finding mistakes using, 349

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

geometric sequences, formula for, 282–284, 285–286
inequality proven with, 293–295
method of proof by, 278
principle of, 277–278
property of a sequence proven with, 295–296
proving formulas, 275–286
strong, 301–308
summation of first n integers, formula for, 278–282
trominoes and, 296–297
Mathematical structures, 867–868
Matrix(ces)
2 x 2, 705–706
adjacency, 699–700, 701
connected components and, 702–703
definition of, 698
directed graphs and, 699–701
graph representations of, 698–710
identity, 706–707
main diagonal of, 698–699
multiplication, 703–707
powers of, 707
products of, 704–705
square, 698
symmetric, 701–702
terminology, 699
undirected graphs and, 701–702
Maurolico, Francesco, 277
Maximal elements, 554–555
McCarthy, John, 372
McCarthy's 91 function, 372
McCulloch, Warren S., 828
Memory dump, reading, 106
Merge sort algorithm, 214, 820–824
Mersenne, Marin, 232
Mersenne primes, 232
Message authentication, 542–543
Method
collision resolution, 444
complete enumeration, 618
of direct proof, 167
of exhaustion, 111, 166
of generalizing from the generic particular, 166–167, 184–185
of iteration, 340–345
of proof by contradiction, 218–221
of proof by contraposition, 222–224
of proof by mathematical induction, 278
Microprocessor, 80
Milhailescu, Preda, 232
Mill, John Stuart, 146
Minimal elements, 554–555
Minimum spanning trees, 744–751
Minor premise, 68
mod/modulo, 201–205
computing, 216
congruence modulo 2 relation, 488
congruence modulo 3 relation, 500–501, 516–517
congruence modulo *n*, 516–518
inverse modulo *n*, 534–536
Modular arithmetic, 528–531
computing product modulo *n*, 530
getting started with, 529
Modular equivalence, 526–527
Modus ponens, 68–69, 76
universal instantiation and, 147–149
Modus tollens, 68–69, 76
universal, 149–150
Monty Hall problem, 567–568
Multigraphs, 681
Multiple
of function, 764
least common, 256
Multiple-input AND-gate, 86–87
Multiple-input OR-gate, 86
Multiplication, A–1
matrix, 703–707
Multiplication rule, 574–584
difficult or impossible to apply, 578–579
subtle use of, 579
Multiplicative identity, 235, 706
Mutually disjoint sets, 385

NAND-gates, 89–90
Napier, John, 801
n-ary relations, 487, 491–492
National Security Agency, 524
Natural logarithms, 451
Natural numbers, 7–8
Naur, Peter, 722
"*n* choose *r*" notation, 269–270
Necessary conditions, 61–62, 128–129
converting to if-then form, 62
interpreting, 61
Negation, 40
of *and* and *or*, 45–46
of conditional statement, 56–57
De Morgan's laws, 45–48
double negative property and, 44–45
of *if-then* statements, 57
of quantified statements, 122–124
of statements with more than one quantifier, 137–138
of universal conditional statements, 124–125
of universal statements, 122
Negative numbers, A–3
Neither-nor, 39–40
Nelson, Edward, 34
Nested loops
algorithm with, 790–791
number of iterations of, 578
.NET computer language, 836
Neurons, 828
Newton, Isaac, 151
Next-state function, 843
Next-state tables, 842, 843, 844–845
n! factorial
Node, 681

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Noncomparable elements, 553
Noncomputable functions, 483
Nonconstructive proof of existence, 164
Nondeterministic finite-state automata, 853
Nondivisibility, checking for, 192
Nonequivalence, showing, 45
Nonfunctions, 427
Nonisomorphic trees, 727–729
Nonoverlapping sets, 385
Nonregular languages, 854–855
Non-trees, 721
NOR-gates, 89–90
Notation
  for algorithms, 248
  Backus-Naur, 722, 829
  big-*O*, big-omega, and big-theta, 769–783
  binary, 93–96
  decimal, 93
  exponential, 93
  factorial, 268–270
  formal logical, 139–141
  function, 19
  hexadecimal, 103–106
  inequalities, 40
  infix, 831
  interval, 382–383
  "*n* choose *r*," 269–270
  octal, 107
  Polish, 831–832
  postfix, 831
  prefix, 831
  product, 264
  set-builder, 8–9
  set-roster, 7–8
  summation, 261–267
  for walks, 680
NOT-gate, 82
NP-complete problems, 824
n-tuples, 432, 492
  ordered, 11–12
Null set, 384
Null strings, 13, 366
Numbers. *See* Integers; Rational numbers; Real numbers,
    natural, 7–8
Number systems, 93–96
Number theory, 190
  cryptography and, 543
  divisibility, 190–197
  Euclid's lemma and, 539–540
  floor and ceiling, 211–216
  open questions in, 232–233
  properties of integers, 186–187
  properties of rational numbers, 185–186
  quotient-remainder theorem, 201–205
Numerical expressions, 341

Octal notation, 107
Odd integers, 161–163
  properties of, 186–187
  square of, 205–207
Of order at least *g*, 769, 771, 773
Of order at most *g*, 770, 771, 773
Of order *g*, 770, 771, 773
Of order *g*(*n*), 789
Omega-notation, 769–783
  description of, 769–771
  polynomial function orders and, 773–774
  for polynomial with negative coefficients, 774–777
  reciprocal relationship between *O*-notation and, 782
  translating to, 772
One-dimensional arrays, 270
  counting elements of, 569–570
One-to-one correspondences, 451–454, 473
One-to-one functions, 439–445
  composition of, 466–468
  finite sets and, 613–614
  finite sets defining, 441
  on infinite sets, 441–443
  proving or disproving, 442–443
One-way functions, 445
Only-if, 162
Only if statements, 128–129
  and the biconditional, 59–61
  converting to if-then, 59–60
O-notation, 769–783
  caution about, 778–779
  description of, 769–771
  polynomial function orders and, 773–774
  for polynomial with negative coefficients, 774–777
  reciprocal relationship between omega-notation and, 782
  showing one function is not big-*O* of other function, 779–780
  translating to, 772
Onto functions, 445–449
  composition of, 468–470
  finite sets and, 613–614
  finite sets defining, 446–447
  incorrect proof, 470
  on infinite sets, 447–449
  proving or disproving, 447–449
Open sentences, 108
Operations
  order of, 39, 55, 60
  on sets, 381–384
Order, of quantifiers, 138–139
Order axioms, A–2
Ordered *n*-tuples, 11–12
Ordered pairs, 10–11, 760
  equivalence clases of relation given as set of, 511
Ordered selections, 618
Ordered triples, 11
Order of operations, 39, 55
  for logical operators, 60
Order of precedence, 833
Ordinal numbers, 473
OR-gate, 82
  multiple-input, 86

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Origin, 8, 760
Or statements, 40, 41
  ambiguity and, 41
  negation of, 45–48
Output signals, 81

Pairwise disjoint sets, 385
Pairwise relatively prime integers, 534
Palindrome, 830
Parallel, switches in, 80
Parallel adders, 99
Parallel edges, 25
Parent, 733, 734
Parentheses, 733
  property of the set of, 369
  recursive definition of, 365–366
Parity, of integers, 203–204
Parse trees, 722–723
Partially ordered sets, 553–555
Partial order relations, 546–560
  applications, 557–560
  definition of, 548
  Hasse diagrams for, 550–553
  lexicographic order, 549–553
  partially and totally ordered sets, 553–555
  PERT and CPM for, 558–560
  subset of, 548
  topological sorting, 555–557
Partitions
  relation induced by, 506–508, 514–515
  of sets, 384–386
Pascal, Blaise, 183, 277, 568, 643
Pascal's formula, 642–646
  new formula derivation from, 646
Pascal's triangle, 644–645
Passwords, with 3-5 letters, 592–593
Paths, 677–692
Peano, Giuseppe, 382, 520
Peirce, Charles Sanders, 16, 89, 110
Peirce arrow, 89–90
Permutations, 580–584
  definition of, 580
  of letters in word, 580–581
  of objects around circle, 581
  relation between combinations and, 618–620
  r-permutations, 581–583
  of selected letters of word, 583–584
  of selected objects, 581–584
  with sets of indistinguishable objects, 629
  of set with repeated elements, 628–629
Personal identification numbers (PINs)
  counting, 575–576
  with repeated symbols, counting, 590–591
PERT (Program Evaluation and Review Technique), 558–560
Piaget, Jean, 53
Pigeonhole principle, 604–614
  applications of, 605–612
  generalized principle of, 610–612
  proof of, 612–614
PINs. *See* Personal identification numbers
Pitts, Walter, 828
Plaintext, 524
Poker hand problems, 626–627
Polish notation, 831–832
Polyá, George, 6
Polynomial functions
  big-theta for, 777–778
  with negative coefficients, 774–777
  orders of, 773–783
Polynomial-time algorithms, 824
Polyomino, 296
Positive closure of Σ, 830
Positive integers, "divides" relation on set of, 548–549
Positive real numbers, A–2
Possibility trees, 573–584
Post, Emil, 828
Post-conditions
  algorithm, 315–316
  correctness of, 317, 319, 320–321, 322–323
  loop, 316–317
Postfix notation, 831
Power functions, 761–762
  orders of, 773
Powers
  of adjacency matrix, 699–700
  of matrix, 707
Power set axiom, 386
Power sets, 386
  function defined on, 429–430
  function from, to set of strings, 452
  number of elements in, 369
  relations on, 488
Pre-conditions
  algorithm, 315–316
  loop, 316–317
Predicate calculus, 108
Predicates
  definition of, 108–109
  finding truth values of, 109
  quantified statements and, 108–129
Prefix notation, 831
Premises, 37, 66
  ambiguous, 72
  false, 74
  major, 68
  minor, 68
  truth set of, 109
Prime integers, 163–164
Prime numbers, 163–164
  divisibility and, 192
  divisibility by, 194, 302–303
  Fermat, 232
  infinite number of, 230–231
  Mersenne, 232
  twin primes conjecture, 232

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Prim's algorithm, 748–751
Principle of mathematical induction, 277–278
Probability
  Bayes' theorem and, 666–668
  binomial, 672
  of complement of event, 656
  of complement of event, formula for, 591
  conditional, 662–665
  counting and, 564–570
  for deck of cards, 566–567
  dice roll, 567
  equally likely probability formula, 566
  expected value and, 658–660
  of general union of two events, 657–658
  independent events and, 668–672
  introduction to, 564–570
  Monty Hall problem, 567–568
  pigeonhole principle and, 604–614
  possibility trees and multiplication rule, 573–584
Probability axioms, 655–660
Probability function, 655
Problems for the Quickening of the Mind, 36
Problem solving
  with recursion, 328
  strategy, 409
  using proof, 224–225
Processors, 842–843
Productions, 722
Product modulo n, 530
Product notation, 264
Products, A–1
  Cartesian, 10–13, 430, 576–577
  computing, 264, 305
  correctness of loop to compute, 318–319
  dot, 703
  of matrix, 704–705
  properties of, 265
  recursive definition of, 335–336
  scalar, 703
Program Evaluation and Review Technique (PERT), 558–560
Programming languages. *See* Computer languages
Prolog, 141–142
Proof(s)
  See also Algebraic proof; Direct proof; Disproof
  algebraic, of set identities, 410–412
  of binomial theorem, 648–651
  common mistakes, 175–178
  for conditional statements, 402
  conjecture and, 179–180
  by contradiction, 218–221, 223–224
  by contraposition, 222–224
  defining, 160–161
  from definitions, 186
  of De Morrgan's law for sets, 398–400
  of distributive law, 395–398
  of divisibility, 291–293, 302–303
  by division into cases, 71, 76, 204–205
  of double complement law, 417–418
  of equality, 284–285
  of existential statements, 164
  identifying mistakes in, 179
  indirect, when to use, 231–232
  of inequality, 293–295
  by mathematical induction, 278
  method of exhaustion, 166
  of number of multiplications needed to multiply n numbers, 305–308
  of pigeonhole principle, 612–614
  as problem-solving tool, 224–225
  for properties of divisibility, 192–194
  of properties of rational numbers, 185–186
  of property of floor, 213–216
  of property of sequence, 295–296, 303–304
  of reflexivity, 500
  of square of odd integer, 205–207
  starting, 170–171
  of transitivity, 501
  of universal statements, 165–170, 173–175
  use of universal modus ponens in, 148–149
  variations among, 175
  writing advice, 173–180
Proper subsets, 9, 378
Propositional calculus, 108
Propositional forms, 42
Propositional functions, 108
Propositions, 222–223
Pseudocode, 244
Public key, 537
Public-key cryptography, 525–526
Pythagoras, 228
Pythagoren theorem, 228
Python language, 201, 244, 593–594, 836

QED, 836
Quantified statements, 108–152
  arguments with, 146–156
  formal vs. informal language and, 112–113
  implicit, 116–117
  logical equivalence for, 122–123, 125–126
  negations of, 122–124
  predicates and, 108–129
  proving validity of arguments with, 150
Quantifiers
  bound variables and scope, 115–116
  definition of, 109–110
  existential, 111–112
  order of, 138–139
  statements with multiple, 131–142
  trailing, 113
  universal, 109–111
Quaternary relations, 491
Quine, Willard Van Orman, 55
Quotient automaton, 859, 863–866
Quotient-remainder theorem, 200–201, 203, 204–205, 215, 271, 527
  well-ordering principle and, 309–310

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Ralston, Anthony, 289
Random process, 565
Range, 426
Rational numbers
  definition for, 519–520
  definition of, 183
  determining irrational numbers vs., 183–184
  direct proof and counterexample with, 183–187
  double of, 187
  as equivalence classes, 519–520
  properties of, 185–186
  set of all positive, 477–478
  sum of irrational and, 220–221
  sum or rationals is rational, 185–186
r-combinations, with repetition allowed, 634–639
Real numbers
  cardinality of set of all, 481–482
  floor and ceiling of, 211–216
  less-than relations for, 487
  negative, A–3
  no smallest positive, 135
  positive, A–2
  properties of, A–1, A–2, A–3
  set of, as uncountable, 478–480
Real-valued functions of real variable, 760–766
Reasoning, circular, 72
Reciprocals, A–1
Recognizer, 85
Recurrence relations, 325
  characteristic equation of, 354–356
  for sequence, 817–820
  initial conditions for, 325
  iterations solving, 340–349
  logarithms solving, 804–806
  second-order linear homogeneous, with constant coefficients, 352–362
  sequences satisfying, 326–327
  solution to, 340
  using mathematical induction to verify correctness of solution to, 347–348
  writing in more than one way, 326
Recursion
  compound interest and, 333–335
  computing number of edges of K_n using, 346–347
  Fibonacci numbers and, 332–333
  recursively defined sequences, 325–337
  Tower of Hanoi and, 328–331
Recursive definition
  of Boolean expressions, 365
  general, 364–372
  of parenthesis structures, 365–366
  of product, 335–336
  of sets, 365–372
  of sets, structural inductions, 367–368
  for sets of strings, 366–367
  of sum, 335–336
Recursive functions, 372–374
Recursive leap of faith, 328
Recursive paradigm, 328
Reductio ad absurdum, 218
Reductio ad impossible, 218
Reflexive property of cardinality, 473
Reflexivity, 495–502
  proof of, 500
Regular expressions, 829
  for date, 838
  definition of, 832
  finite-state automata and, 851–853
  language defined by, 832–836
  order of precedence for operations in, 833
  practical uses of, 836–838
  symbols in, 833
Regular languages, 828–829, 854–855
Relational databases, 491–492
Relations
  See also "Divides" relations; Equivalence relations; Recurrence relations
  antisymmetry property of, 546–548
  arrow diagram of, 18
  binary, 487, 491
  circle, 17
  congruence modulo 2, 488
  congruence modulo 3, 500–501, 516–517
  congruence modulo *n*, 516–518
  directed graph of, 490–491, 495
  of equality, 498–499
  equivalence, 505–520
  finite, 489, 547
  on finite sets, 19–20
  finite sets and, properties of, 496–498
  identity, equivalence classes of, 512–515
  infinite, 490
  infinite sets and, properties of, 498–501
  inverse of, 488–490
  language of, 15–18
  less-than, 499
  less than, for real numbers, 487
  "less than or equal to," 549
  n-ary, 487, 491–492
  partial order, 546–560
  partition inducing, 506–508, 514–515
  on power set, 488
  quaternary, 491
  recurrence, 804–806
  reflexivity, symmetry, and transitivity and, 495–502
  on set of identifiers, 510
  on set of subsets, 511–512
  on sets, 487–492
  as subsets, 16–17
  ternary, 491
  total order, 553
  transitive closure of, 502
Relative complement, 381
Relatively prime integers, 534–535
Repeating decimals, 184
Repetition, *r*-combinations with, 634–639

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Residues modulo *n*, 527
Reverse Polish notation, 831
Ribet, Kenneth, 180
Right child, 734
Right subtree, 734
Ritchie, Dennis, 829*n*
Rivest, Ronald, 525–526
Rooted trees, 732–741
Roussel, P., 141
Rows, multiplying, 703–704
r-permutations, 581–583
RSA cipher, 525–526
RSA cryptography, 536–543
Rules of inference, 69–72, 76
Russell, Bertrand, 301, 340, 419, 800
Russell's paradox, 419–420

Sample space, 565–566
Saturated hydrocarbon molecules, 723–724
Savage, Carla, 394
Sawyer, W. W., 677
Scalar product, 703
Schroeder-Bernstein theorem, 486
Scope, bound variables and, 115–116
Search algorithm
  binary, 214, 813–825
  sequential, 787–788
Second-order linear homogeneous recurrence relations, with constant coefficients, 352–362
Semantics, 723
Sequences, 258–273
  See also Recursion
  alternating, 260
  arithmetic, 342–343
  of Catalan numbers, 327
  in computer programming, 270–271
  defining recursively, 325–337
  definition of, 259
  definition of limit of, 135–136
  explicit formula for, 259–261
  factorial notation and, 268–270
  Fibonacci, 358–360
  finite, 270
  floor function and, 304–305
  functions and relations on, 429
  general formula for, 259
  geometric, 282–284, 285–286, 343–345
  infinite, 259
  product notation and, 264
  proving properties of, 295–296, 303–304
  recurrence relation for, 817–820
  summation notation and, 261–267
  terms in, 259
  Tower of Hanoi, 328–331, 345–346
  Venn diagrams and, 380–381
Sequential circuits, 83, 841
Sequential search algorithm, 787–788
Set difference law, 395
Set difference property, deriving, 411
Set identities, 394–400
  algebraic proofs of, 410–412
  counterexample for, 407–409
  deriving, 411–412
Set notation
  describing language defined by regular expression, 833–834
  set-builder notation, 8–9
  set-roster notation, 7–8
Set(s)
  algorithm for checking for subsets of, 386–387
  axiom of extension, 7
  Boolean algebra and, 414–418
  Cartesian products and, 10–13
  complements of, 381, 382
  countable, 475–477
  definitions, procedural versions, 392
  De Morgan's law for, 398–400
  differences of, 381, 382
  disjoint, 384–385
  disjoint, counting elements of, 589–599
  disproving alleged property, 407–409
  distributive law for, 395–398
  empty, 384, 401–403
  equality, 379–380
  finite, 473, 612–614
  finite, composition of functions defined on, 463
  finite, functions and relations on, 19–20, 441, 446–447
  finite, properties of relations on, 496–498
  functions acting on, 434–435
  functions defined on general, 425–435
  of identifiers, 510
  inclusion/exclusion rule, 595–599
  indexed collection of, 383
  infinite, 473, 474–475, 612
  infinite, one-to-one functions on, 441–443
  infinite, onto functions on, 447–449
  infinite, properties of relations on, 498–501
  of integers, 368–369
  of integers, graphing functions defined on, 763
  intersection of, 381, 382, 383–384
  language of, 6–15
  mutually disjoint, 385
  nonoverlapping, 385
  null, 384
  of numbers, relations among, 381
  operations on, 381–384
  ordered pairs and, 10–11
  pairwise disjoint, 385
  of parentheses, 369
  partially and totally ordered, 553–555
  partitions of, 384–386, 506–508
  power, 386
  power sets of, 386
  properties of, 367–372, 391–403
  recursively defined, 365–372
  relations and, 15–20, 487–492

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Russell's paradox and, 419–420
of strings, 452
of strings, functions and relations on, 20
subsets, 9–10, 378–379, 617–630
subsets of, number of, 409–410
uncountable, 477–481
union of, 381, 382, 383–384
universal, 381
Venn diagrams and, 394
Set theory, 377–424
Seven Bridges of Königsberg, 677–678
Shakespeare, William, 122
Shamir, Adi, 525–526
Shannon, Claude, 79, 828
Sheffer, H. M., 89
Sheffer stroke, 89–90
Shortest path algorithm, 742–757
Sieve of Eratosthenes, 227
Sigma, 829–830
Signed integers, computer representation of, 99–101
Simple circuit, 679–680
Simple graphs, 239–240
isomorphisms of, 717–718
Simple vending machine, 841–842
Singh, Simon, 180*n*
Single-root theorem, 360–362
Smullyan, Raymond, 75
Software, simulating finite-state automaton using, 849–851
Solution, to recurrence relation, 340
Sorting
insertion algorithm for, 793–796
merge sort algorithm for, 214, 820–824
topological, 555–557
Sound arguments, 74, 150
Space efficiency of algorithm, 788, 825
Spanning trees, 742–757
definition of, 743
Dijkstra's shortest path algorithm and, 751–757
Kruskal's algorithm and, 745–748
minimum, 744–751
Prim's algorithm and, 748–751
Specializations, 70, 76
Square, of odd integer, 205–207
Square matrix, 698
Square roots, irrationality of square root of two, 228–230
Squaring function, 21
Standard factored form, 196
Star *-equivalence classes, 863
Statement calculus, 108
Statement forms, 42
Statements. *See* also *specific statements*
combinations of, 3–5
conditional, 2
definition of, 38–39
existential, 2
formal logical notation and, 139–141
logically equivalent, 44
with multiple quantifiers, 131–142
negation of, with more than one quantifier, 137–138
translating from informal to formal language, 134–136
types of, 2
universal, 2, 110
vacuously true, 126
States, 843
(state-) transition diagram, 843–844
Stevin, Simon, 478
String-reversing function, 452–453
Strings, 13
additive property of length, 370
area code, 857
bit, 13
characters as, 366
characters of, 13
concatenation, 20, 366, 370–372
in language defined by regular expressions, 834–835
length of, 369–370
local call, 857
null, 13, 366
properties of, 367
recursive definition of sets of, 366–367
set of, function from power set to, 452
sets of, functions and relations on, 20
testing for lexicographic order, 550
Strong mathematical induction, 301–308
applying, 302–308
computation of products with, 305
divisibility by a prime proven by, 302–303
number of multiplications needed to multiply *n* numbers proven with, 305–308
principle of, 301–302
property of a sequence proven with, 303–304
Structural induction, 364, 367–372
Subgraphs, 681–682
Sublist, counting elements of, 569
Subroutines, 248
Subscript, 259
"Subset" relation, 548
Subsets, 9–10
algorithm for checking for, 386–387
chain of, 554
counting, 617–630
intersection of, 400
number of, of set, 409–410
proof and disproof, 378–379
proper, 9, 378
proving relations, 392–394
relations, 378–379, 391–394
relations as, 16–17
testing, 378
union of, 400
Subtraction
in binary notation, 96
integers, in two's complement form, 102–103
odd minus even integer, 178–179
Subtrees, 734
Successor function, 21

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Sufficient conditions, 61–62, 128–129
converting to if-then form, 61
interpreting, 61
Sum
harmonic, 809–810
of rational and irrational number, 220–221
of rational numbers is rational, 185–186
recursive definition of, 335–336
simplifying, using binomial theorem, 652
telescoping, 264
Summands, 182
Summations
change of variable and, 266–267
in closed form, 281
computing, 261
evaluating, 263
expanded form of, 261–262
of first n integers, formula for, 278–282
geometric sequences, formula for, 282–284, 285–286
notation of, 261–267
properties of, 265
telescoping sums, 264
upper limit of, 261, 267
Sum-of-products form, 88
Surjective (onto) functions, 445–449
Swift, Jonathan, 325
Switches
in parallel, 80
in series, 80
Syllogisms, 68–69
Symbolic logic, 37
Symmetric matrix, 701–702
Symmetric property of cardinality, 473
Symmetry, 495–502
Syntactic derivation trees, 722–723
Syntax, 723
Syntax errors, 314

Tables
input/output, 81, 87–88
next-state, 842, 843, 844–845
trace, 246–247
truth, 41–43
Taniyama-Shimura conjecture, 180
Tao, Terence Chi-Shen, 232
Tarski, Alfred, 118
Tarski's World, 117–118, 132–133
evaluating argument for, 154–155
formalizing statements in, 139–140
negating statements in, 137–138
quantifier order in, 139
Tautological statements, 48–49
Tautologies
definition of, 48
logical equivalence and, 49
Taylor, Richard, 180
Teams
calculating number of, 620
with members of two types, 623–625
that contain both or neither, 620–621
that do not contain both, 621–623
Telescoping sums, 264
Terminal vertex, 725
Terms, in sequences, 259
Ternary relations, 491
Theorem. *See* also *specific theorems*
definition of, 169
direct proof of, 168–170
There exists statement, 125
Theta-notation, 769–783
deducing, 772
description of, 769–771
for polynomial function, 777–778
polynomial orders and, 780
reflexive, symmetric, and transitive properties of, 782
translating to, 771–772
Thinking Machines Corporation, 180
Thompson, Kenneth, 829*n*
Thoreau, Henry David, 858
$3n + 1$ problem, 374
$3x + 1$ problem, 374
Time efficiency, of algorithm, 788–789, 825
Topological sorting, 555–557
Totally ordered sets, 553–555
Total weight, 745
Tournament play, possibilities for, 573–574
Tower of Hanoi, 328–331, 345–346, 824
Trace tables, 246–248
for division algorithm, 249–250
for insertion sort, 793–794
Tractable problems, 824
Trailing quantifiers, 113
Trails, 679–681
Euler, 688–689
Transition diagrams, 843–844
Transitive closure, of relation, 502
Transitive law, A–3
Transitive property of cardinality, 474
Transitivity, 71, 76
of divisibility, 192–194
proof of, 501
relations and, 495–502
Traveling salesman problem, 692
Tree(s), 720–759
binary, 734–739
binary search, 739–741
characterizing, 724–730
conditional probabilities represented with, 663–665
decision, 721–722
Dijkstra's shortest path algorithm and, 751–757
examples of, 720–724
graph, 720, 727
Kruskal's algorithm and, 745–748
nonisomorphic, 727–729
non-trees and, 721
parse, 722–723

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Prim's algorithm and, 748–751
rooted, 732–741
satisfying given conditions, 727
spanning, 742–757
subtrees, 734
syntactic derivation, 722–723
theorems about, 725–727, 729–730
vertices and, 724–727
Trefethen, Lloyd, 566
Trefethen, Nick, 566
Triangle inequality, 207–208
Trichotomy law, A–3
Triples, counting, 637
Trominoes, 296–297
True by default, 126
Truth set, of predicates, 109
Truth tables
for biconditional, 60
for compound statements, 42–43
for conditional statements, 55
for conjunction, 41
for disjunction, 42
for exclusive *or*, 42–43
for negation, 40–41
Truth values, 40–42
of predicates, 109
Tucker, Alan, 634
Tukey, John W., 80–81
Turing, Alan M., 420, 787, 828, 843
Turing machine, 828
Twin primes conjecture, 232
Two-dimensional Cartesian coordinate system, 760
Two's complements
8-bit, 99–103
addition/subtraction with integers and, 102–103
computer representation of signed integers and, 99–101
finding, 100–101
finding number with given, 101
for negative integer, 100

Uncountable sets, 477–481
Undirected graphs, matrices and, 701–702
Union
counting elements of general, 596–597
interaction of function with, 434–435
of mutually disjoint subsets, 385
of sets, 381, 382, 383–384
with subset, 400
of two events, probability of, 657–658
Unique factorization of integers theorem, 195–197, 251
Euclid's lemma and, 539–540
Uniqueness of the complement law, 417
Universal bound laws, 395, 416
Universal conditional statements, 113–114
contrapositive, converse, and inverse of, 127–128
definition of, 3
necessary and sufficient conditions and, 128
negations of, 124–125
only if and, 128–129
rewriting, 3
variants of, 126–128
writing informally/formally, 113–114
Universal existential statements
definition of, 3–4
rewriting, 4
Universal generalization, 166
Universal instantiation, 146–147, 150
Universal modus ponens, 147
drawing conclusions using, 148
recognizing, 147–148
use of, in a proof, 148–149
Universal modus tollens, 149
drawing conclusions using, 150
recognizing, 149
Universal quantifiers, 109–111
Universal set, 381
Universal statements, 2
definition of, 110
disproving by counterexample, 164–165
equivalent forms of, 114–115
existential, 3–5
negation of, 122
proofs of, 173–175
proving, 165–170
truth and falsity of, 110–111
vacuous truth of, 126
Universe of discourse, 381
UNIX, 829, 836
Unordered selections, 618
Unsound arguments, 74

Vacuously true statements, 54, 126
Valid arguments. *See* Arguments, valid and invalid
Validity, using diagrams to test for, 151–155
Vandermonde, Alexander, 654
Vandermonde convolution, 654
Variables
in algorithmic language, 244
Boolean, 84
bound, 115–116
change of, 265–267
in computer languages, 244
data type of, 244
dummy, 266, 270–271
function of two, 453–454
global, 176
uses of, 1–2
writing sentences using, 2
Vending machines, 841–842
Venn, John, 380
Venn diagrams, 380–381, 394
Vertex (vertices), 24–25, 681
adjacent, 25
branch, 725
degree of, 31–32
internal, 725–727

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203

Vertex (vertices) (*Continued*)
isolated, 25
terminal, 725
Vertical axes, 760
Volterra, Vito, 425

Walks, 680–681
definition of, 679–680
of length N, counting, 707–710
notation for, 680
Weighted graphs, 745–748
Well-defined functions, 433–434
Well-ordering principle, 258, 277
for integers, 308–310
Weyl, Hermann, 720
Wheeler, Anna Pell, 200, 439, 573
While loops, 245–246, 316, 793–794
guard of, 320
tracing execution of, 246–247
Whitehead, Alfred North, 24, 461, 732
Wiener, Norbert, 10, 841
Wiles, Andrew, 180
World Wide Web, 28–29
Worst-case orders, 824
for sequential search, 787–788

XML, 829

Zero, divisors of, 190–191
Zero factorial (0!), 269
Zero product property, 184
Zhang, Yitang, 232

Copyright 2020 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part. WCN 02-200-203